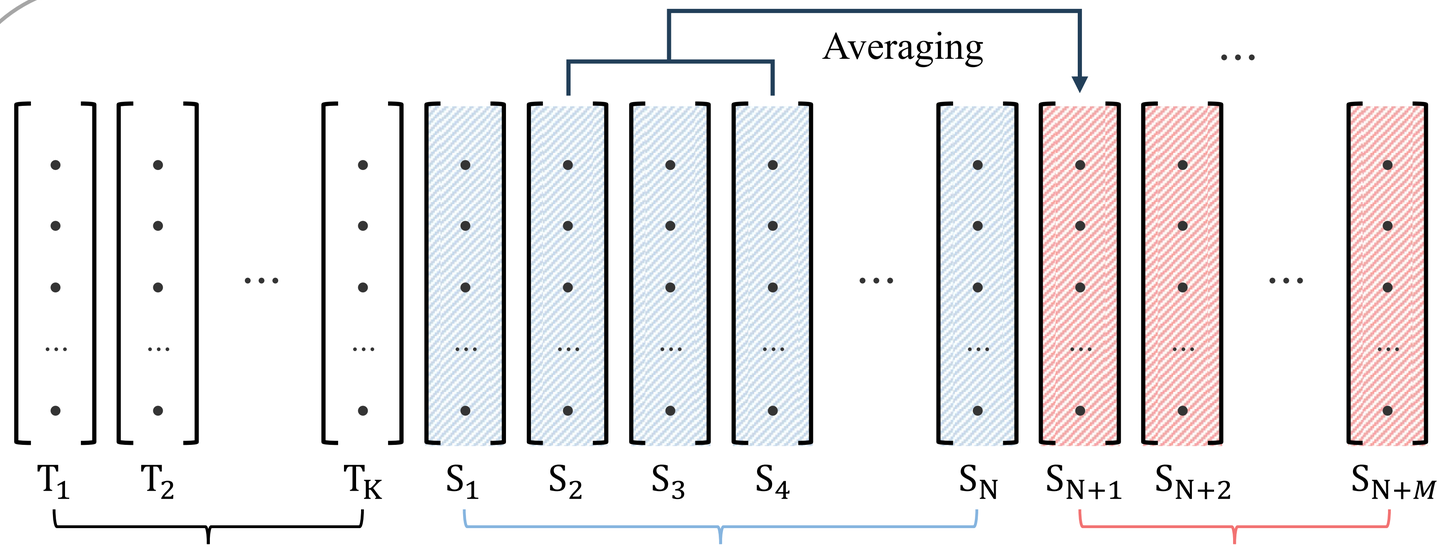
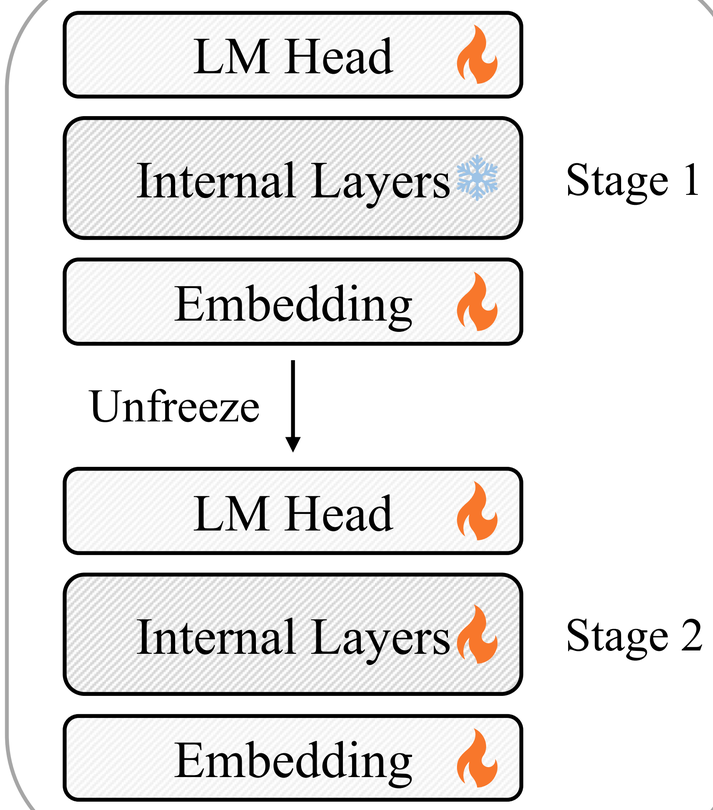
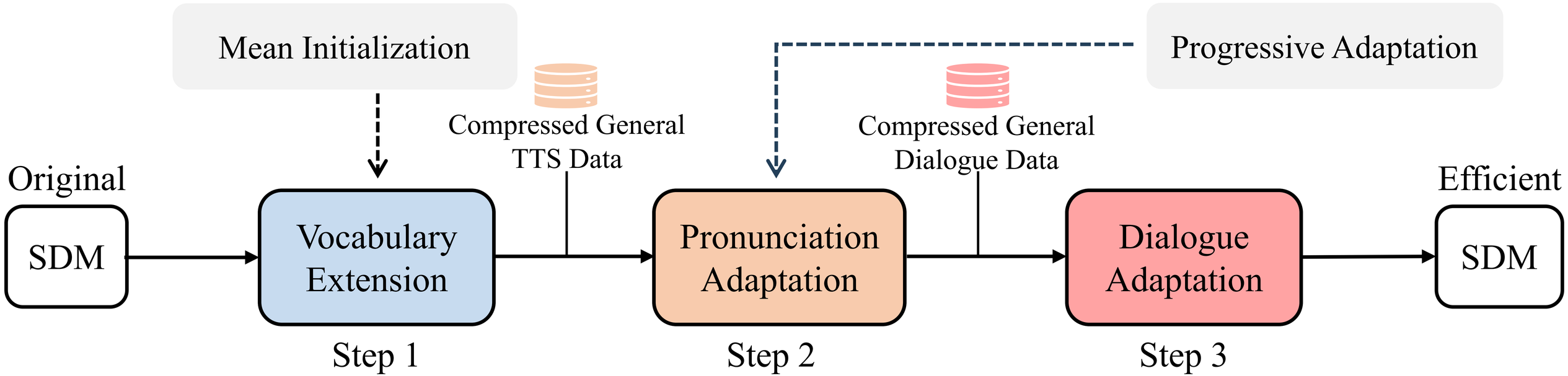

Averaging
T_1 T_2 T_K S_1 S_2 S_3 S_4 S_N S_{N+1} S_{N+2} S_{N+M}
Text Embedding
(Original Vocabulary)
Speech Embedding
(Original Vocabulary)
Extended Speech Embedding
(Newly Added Vocabulary)
Initialized by averaging original-token expansion; same for LM head.


LM Head
Internal Layers
Embedding
Stage 1
Unfreeze
LM Head
Internal Layers
Embedding
Stage 2


Mean Initialization
Progressive Adaptation
Compressed General
TTS Data
Compressed General
Dialogue Data
Original
SDM
Vocabulary
Extension
Pronunciation
Adaptation
Dialogue
Adaptation
Efficient
SDM
Step 1
Step 2
Step 3